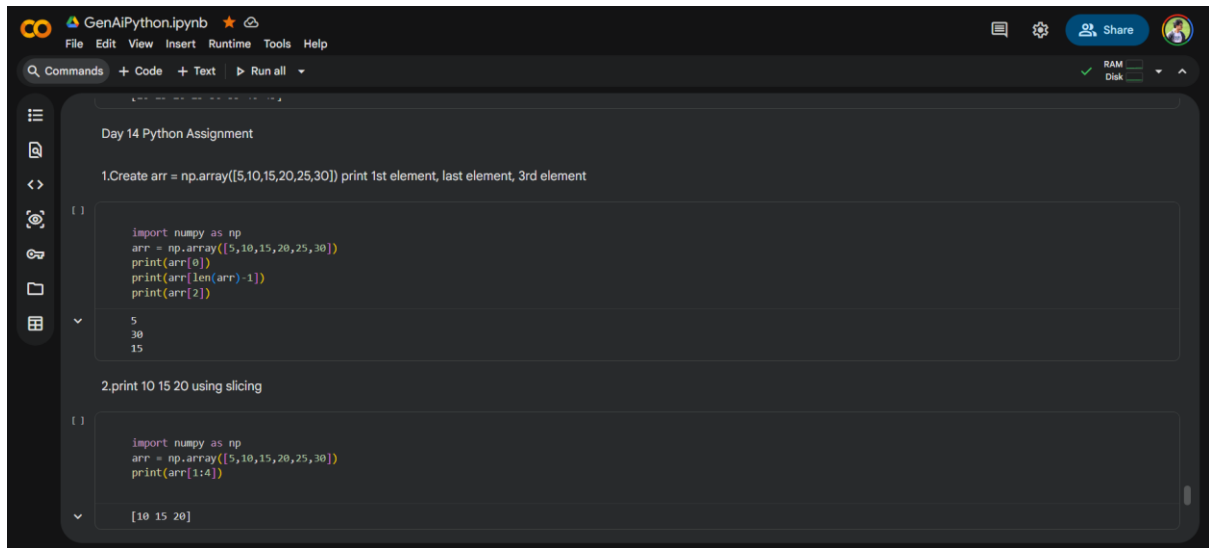


Day 14 Python Assignment



Day 14 Python Assignment

1. Create arr = np.array([5,10,15,20,25,30]) print 1st element, last element, 3rd element

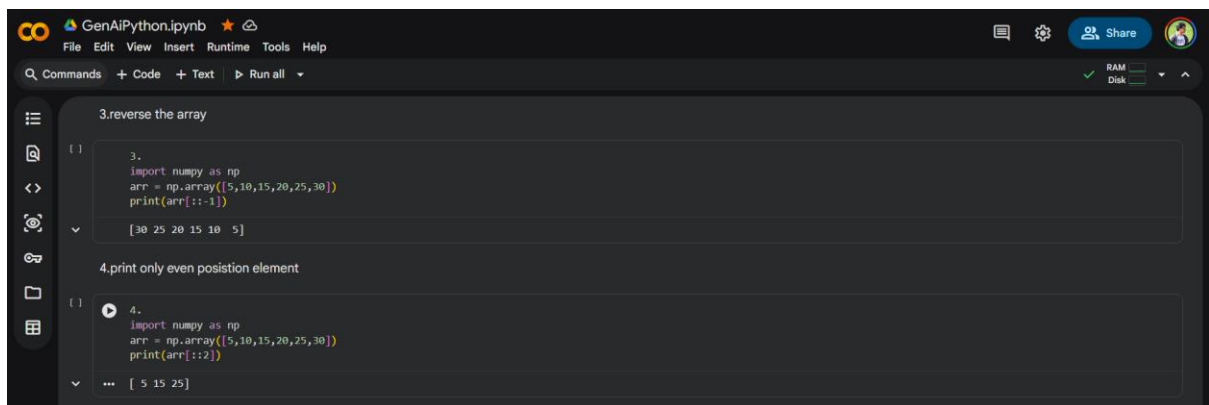
```
[ ]
import numpy as np
arr = np.array([5,10,15,20,25,30])
print(arr[0])
print(arr[len(arr)-1])
print(arr[2])
```

5
30
15

2. print 10 15 20 using slicing

```
[ ]
import numpy as np
arr = np.array([5,10,15,20,25,30])
print(arr[1:4])
```

[10 15 20]



3. reverse the array

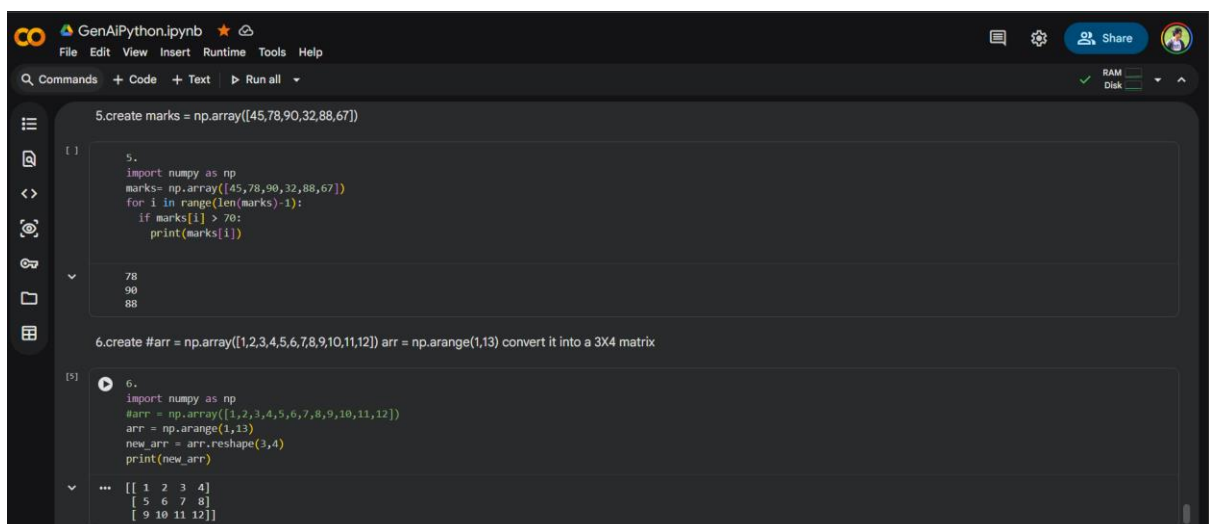
```
[ ]
3.
import numpy as np
arr = np.array([5,10,15,20,25,30])
print(arr[::-1])
```

[30 25 20 15 10 5]

4. print only even position element

```
[ ]
4.
import numpy as np
arr = np.array([5,10,15,20,25,30])
print(arr[::2])
```

[5 15 25]



5. create marks = np.array([45,78,90,32,88,67])

```
[ ]
5.
import numpy as np
marks= np.array([45,78,90,32,88,67])
for i in range(len(marks)-1):
    if marks[i] > 70:
        print(marks[i])
```

78
90
88

6. create #arr = np.array([1,2,3,4,5,6,7,8,9,10,11,12]) arr = np.arange(1,13) convert it into a 3X4 matrix

```
[ ]
6.
import numpy as np
#arr = np.array([1,2,3,4,5,6,7,8,9,10,11,12])
arr = np.arange(1,13)
new_arr = arr.reshape(3,4)
print(new_arr)
```

[[1 2 3 4]
 [5 6 7 8]
 [9 10 11 12]]

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7. flatten the matrix

```
[6]
import numpy as np
arr = np.array([1,2,3,4,5,6,7,8,9,10,11,12])
new_arr = arr.reshape(3,4)
print(new_arr)
print(new_arr.flatten())
```

```
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]
[ 1  2  3  4  5  6  7  8  9 10 11 12]
```

8. multiple every element by 5

```
[7]
import numpy as np
arr = np.array([1,2,3,4,5,6,7,8,9,10,11,12])
print(arr * 5)
```

```
[ 5 10 15 20 25 30 35 40 45 50 55 60]
```

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9. find sum, mean, median, max, min, std

```
[9]
import numpy as np
arr = np.array([1,2,3,4,5,6,7,8,9,10,11,12])
print("sum")
print(arr.sum())
print("mean")
print(arr.mean())
print("median")
print(np.median(arr))
print("max")
print(arr.max())
print("min")
print(arr.min())
print("standard deviation")
print(arr.std())
```

```
...
sum
78
mean
6.5
median
6.5
max
12
min
1
standard deviation
3.452052529534663
```

GenAI Python.ipynb

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10. create 2 2x2 matrix and perform element wise multiplication and matrix multiplication

```
[14]
import numpy as np
arr = np.array([[1,2],[4,5]])
arr1 = np.array([[7,8],[10,11]])
print("element wise multiplication")
print(arr * arr1)
print("matrix multiplication")
matrix = arr @ arr1
print(matrix)
```

```
element wise multiplication
[[ 7 16]
 [40 55]]
matrix multiplication
[[27 30]
 [78 87]]
```

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11. create a series containing 10 20 30 40 50

```
[15]
import pandas as pd
s = pd.Series([10,20,30,40,50])
print(s)
```

```
0    10
1    20
2    30
3    40
4    50
dtype: int64
```

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12.create a series with custom index apple orange banana

[16] ✓ Os

12.
import pandas as pd
price = pd.Series([40,50,60],index=["Apple","Orange","Banana"])
print(price)

... Apple 40
Orange 50
Banana 60
dtype: int64

13.create the following dataframe name age city asha rahul priya 21 24 23 chennai bangalore hyderabad

[17] ✓ Os

13.
import pandas as pd
data = {
 "name":["Asha","Rahul","Priya"],
 "Age":[21,30,28],
 "City":["Chennai","Bangalore","Hyderabad"]
}
df =pd.DataFrame(data)
print(df)

	name	Age	City
0	Asha	21	Chennai
1	Rahul	30	Bangalore
2	Priya	28	Hyderabad